Project Proposal

Course "AI5301-Advanced Artificial Intelligence**"**

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AI Voice Sales & Support Agent

Project Proposal — Real-Time AI Voice Booking Agent for Inbound & Outbound Calls

1. Project Overview

This proposal outlines an AI-powered voice agent that answers and conducts phone calls on behalf of a business, holding a natural, low-latency, two-way voice conversation with callers. The agent is purpose-built for service businesses (e.g., home services such as plumbing, electrical, and AC repair) that need to capture booking details from every inbound call without a live receptionist. The system listens to the caller in real time, responds with natural speech, and automatically extracts and stores structured booking information.

2. Problem Statement

- Missed or unanswered calls translate directly into lost bookings and revenue for service businesses.
- Hiring and staffing a 24/7 human receptionist is costly and difficult to scale.
- Manual intake is inconsistent — agents forget questions, mishear details, or fail to log information correctly.
- Existing chatbot solutions don't address phone callers, who still make up a large share of customer inquiries.

3. Proposed Solution

We propose a hybrid real-time voice pipeline that connects telephony (Twilio), a conversational reasoning engine (OpenAI's Realtime API), and a high-fidelity text-to-speech engine (ElevenLabs), orchestrated by a FastAPI backend. A lightweight React web client is included for live demos and internal testing without needing an actual phone call.

Core capabilities:

- Inbound call handling via a Twilio Media Stream, bridged to the AI in real time over WebSockets.
- Guided, step-by-step conversational flow that collects: service type, issue details, customer name, preferred date, and address.
- Server-side voice activity detection (VAD) for natural turn-taking, with barge-in support (caller can interrupt the agent).
- Live transcript and call-status broadcast to a connected dashboard for real-time monitoring.
- Automatic call summary and graceful call termination once booking details are confirmed.

4. Technical Architecture

Layer	Technology
Telephony	Twilio Voice + Media Streams (WebSocket audio bridge)
Conversational AI	OpenAI Realtime API (gpt-realtime-1.5) — speech understanding, reasoning, function calling
Voice Synthesis	ElevenLabs streaming TTS (eleven_turbo_v2_5), with connection pre-warming for low latency
Backend	Python, FastAPI, WebSockets, asyncio (async, event-driven call handling)
Frontend	React 19 + Vite, live transcript view, browser-based test-call client

5. Scope & Deliverables

- FastAPI backend exposing inbound-call, test-call, and WebSocket media/UI endpoints.
- Hybrid audio pipeline bridging OpenAI Realtime output to ElevenLabs speech synthesis.

- Configurable booking script (currently tailored to a home-services use case) collecting five structured data points per call.
- React dashboard for placing test calls and viewing the live transcript.
- Deployment-ready configuration via environment variables (.env) for API keys and base URL.

6. Timeline (Indicative)

Phase	Activities
Week 1	Backend setup, Twilio + OpenAI Realtime integration, basic call bridging
Week 2	ElevenLabs hybrid audio pipeline, booking flow logic, tool/function calling
Week 3	React test dashboard, live transcript streaming, end-to-end testing
Week 4	Production deployment, persistence layer, QA and handover

7. Expected Outcomes

- 24/7 automated call answering with no missed bookings.
- Consistent, structured data capture for every caller.
- Reduced receptionist workload and lower cost-per-call versus human staffing.
- A reusable voice-agent framework that can be extended to other booking or support scripts.